

CS2201 - Data Structures and Algorithms

End Semester Examination - Answer all questions.

- | | | |
|-------------|--|-----|
| 1(a) | Define a stack and list its basic operations. | [2] |
| 1(b) | Explain how a stack is used to evaluate a postfix expression. | [4] |
| 2(a) | Apply the quick sort algorithm to the list 42, 17, 93, 8, 56, 31 and show each partitioning step. | [5] |
| 2(b) | Compare the worst case time complexity of quick sort and merge sort, and justify which one you would choose for sorting a nearly sorted array. | [6] |
| 3(a) | Why is a hash table using separate chaining preferred over linear probing when the load factor is high? | [4] |
| 3(b) | Design a graph based solution to find the shortest delivery route for a courier service across a city, and analyze the time complexity of your solution. | [8] |
| 4(a) | Explain the rotations performed in an AVL tree when it becomes unbalanced after an insertion. | [5] |